Motor imagery on PhysioNet EEGBCI (S70–S109): what is in the signal, and what do the networks learn?

All numbers and figures are produced by `python main.py` (seed 42); choices with alternatives are in DECISIONS.md, predictions (committed before results) in PREDICTIONS.md.

A1 — Data audit

Runs 4, 8, 12 (imagined left/right fist; T1 = left, T2 = right, T0 = rest — PhysioNet EEGMMIDB protocol page, confirmed in the EDF annotations). `audit.csv` lists, per subject, runs, sampling rate, trial counts, event mapping, anomalies and the action taken; the script refuses to run if an anomaly has no explicit decision. Excluded: S88, S92, S100 (128 Hz with a different cue/rest timing — 1.38 s rest leaves no pre-cue baseline). Kept with fixes: dead/noisy channels interpolated in S79, S86, S97, S105; truncated run 8 of S104 and 13 end-of-run trials whose window does not fit dropped. Result: 37 subjects, 827 left / 822 right trials.

A2 — Physiological ground truth

Reference. Every channel is 'electrode minus an undocumented reference'; whatever the reference picks up appears in C3 and C4 alike, and if it lies off the midline or over sensorimotor cortex it can create or cancel a C3–C4 difference. We use a surface Laplacian (CSD), which is reference-free and local. Mu-lateralised ERD is 'present' in 1/37 subjects as recorded, 3 with common average and 5 with CSD (4 labels change). **Method.** Morlet power (4–35 Hz) at C3/C4 and neighbours; per-trial ERD = 10·log10(power 0.5–3.5 s / power –1.5 to –0.5 s), a baseline inside the 4.1 s rest that ends 0.5 s before the cue and starts after the previous trial's beta rebound. LI = mean(ERD_{C3}–ERD_{C4} | left) – mean(… | right) in the mu band (positive = contralateral); one-sided label-permutation test (5000), BH-FDR over subjects. **Result.** Present in 5/37 (S70, S72, S84, S85, S93); 10 at uncorrected p < .05; group median LI +0.48 dB (Wilcoxon p = 0.004). Contralateral ERD is real on average but reliable in few individuals. 'Absent' covers different cases: S082 (below) shows the expected pattern but misses FDR (q = .08), S109 shows weak ERD at C3 whichever hand is imagined, and S080 desynchronises both hemispheres equally.

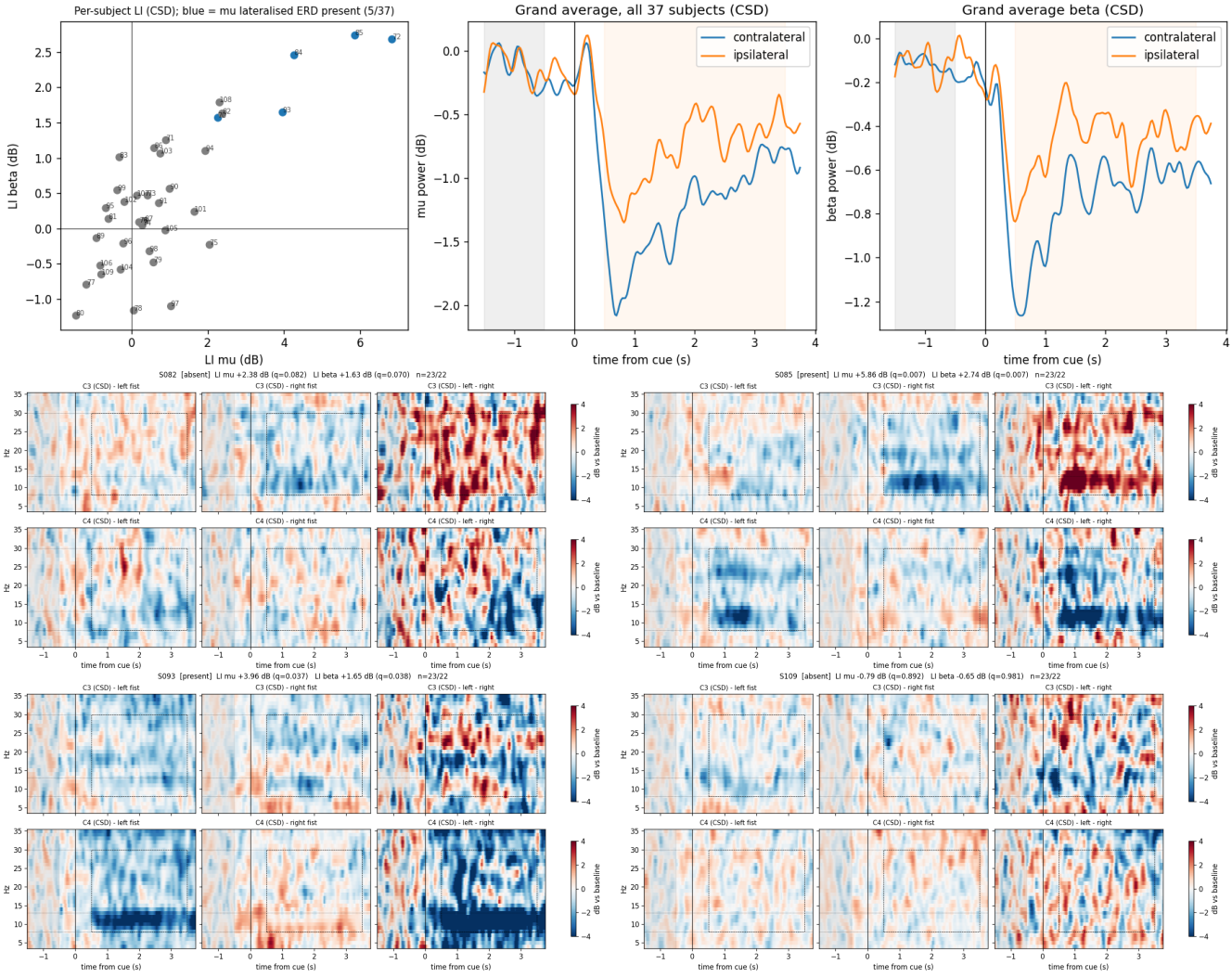


Fig. 1. Top: per-subject mu vs beta LI and grand-average contralateral/ipsilateral ERD (CSD). Bottom: CSD time–frequency maps (dB vs baseline) at C3/C4 for two 'present' and two 'absent' subjects, chosen by rule (strongest/weakest significant; closest to significance/clearest null).

A3 — Confound check

The cue is a target on the *left or right of the screen* for ~4 s, with no documented fixation instruction, so the label is confounded with stimulus position. Routes: (1) **eye movements** towards the target — horizontal-EOG proxies F7–F8, AF7–AF8, Fp1–Fp2, 0.1–8 Hz, 0–3.5 s; (2) **lateralised visual response** — PO7–PO8, PO3–PO4, O1–O2 at 0.05–0.5 s (evoked) and CSD posterior alpha 0.5–3.5 s. Control: CSD sensorimotor mu/beta. Within-subject shrinkage LDA, leave-one-run-out CV. Chance threshold per subject: smallest k with P(X ≥ k | n, 0.5) ≤ .05 (binomial; 64.3–65.1% for n = 42–45); by chance ≈ 1.9/37 subjects pass.

route	mean acc	subjects ≥ threshold	Wilcoxon p (vs 50%)
eye movement (HEOG proxies)	76.0%	31/37	< 0.001
visual evoked	64.5%	20/37	< 0.001
posterior alpha	51.0%	2/37	0.39

motor control (mu/beta, CSD)	52.7%	8/37	0.13
eye, 0.05–0.15 s (post-hoc)	52.8%	4/37	0.033
visual, 0.05–0.15 s (post-hoc)	51.8%	5/37	0.22

Outcome. I predicted the eye route would win but badly underestimated it: subjects look at the target (±45 µV F7–F8 step of opposite polarity from ~0.2 s, Fig. 2). It decodes left/right in 31/37 subjects, including 26 of the 32 A2-absent ones, against 8/37 for the motor route. The 'visual' decoder's peak coincides with the saccade; restricted to the pre-saccadic window it falls to chance, so it is mostly the same eye movement.

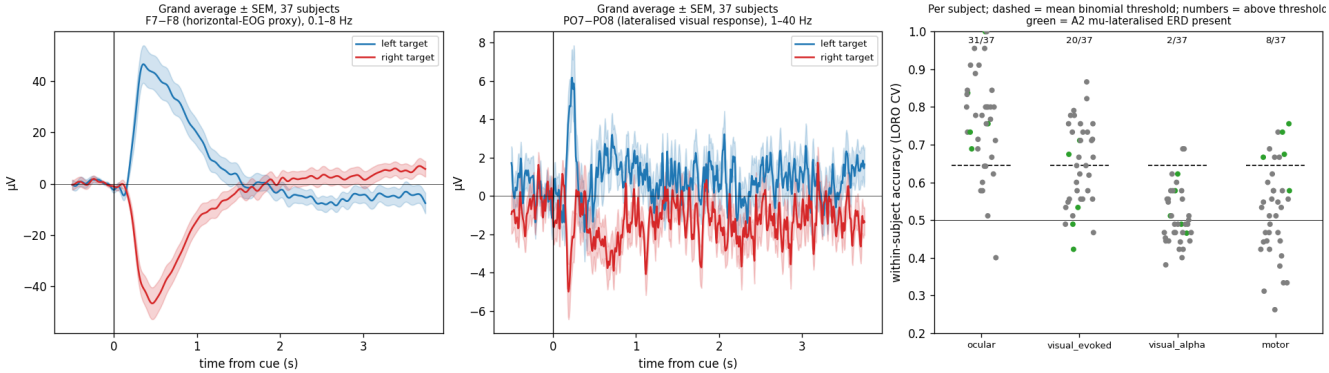
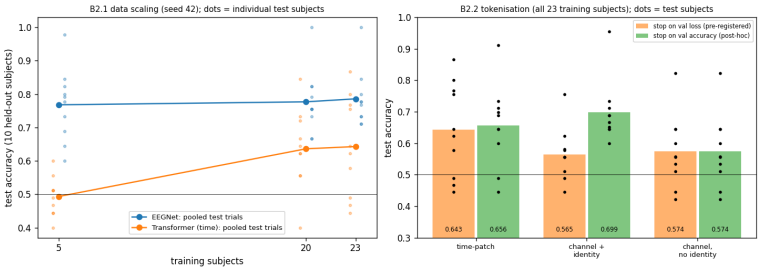


Fig. 2. Grand-average F7–F8 and PO7–PO8 for left/right targets (±SEM, 37 subjects); per-subject accuracies with thresholds (green = A2 present).

B — Networks

Fixed seeded subject split: train 23, validation 4 (early stopping only), test 10 (S74, S75, S77, S89, S95, S97, S98, S99, S105, S109); per-channel z-score from training subjects; input 64 ch × 0–4 s, 1–40 Hz. **EEGNet-8,2** (2962 params) adapted to 160 Hz: temporal kernel 64→80 (0.5 s), separable kernel 16→20 (500 ms at 40 Hz); dropout 0.25 (cross-subject). **Transformer**: tokeniser → 2 pre-norm layers (d = 16) → mean pool; time-patch tokens (spatial projection 64→8, 2 temporal filters per virtual channel, 250 ms patches, 5994 params), channel tokens with/without a learned electrode-identity embedding (6930/5906); all ≤ 2.34× EEGNet. 448 test trials: chance threshold 53.8%, SE ≈ 2.4 points.

run	subj.	val	test
EEGNet	5	71.1%	76.8%
EEGNet	20	72.2%	77.7%
EEGNet	23	71.7%	78.6%
TF time-patch	5	57.2%	49.3%
TF time-patch	20	75.6%	63.6%
TF time-patch	23	70.6%	64.3%
TF channel + id	23	51.7%	56.5%
TF channel, no id	23	61.1%	57.4%
TF time, stop acc*	23	71.7%	65.6%
TF chan + id, stop acc*	23	67.8%	69.9%
* post-hoc (D28)			



B2.1. EEGNet is flat at 76.8/77.7/78.6% for 5/20/all subjects; the transformer rises 49.3 → 63.6 → 64.3%. **B2.2.** With validation-loss stopping (pre-registered) channel + identity stopped at epoch 1 (validation loss rose from the start while accuracy improved); stopping on validation accuracy gives 69.9% vs 65.6% for time patches. Channel tokens without identity: 57.4% (above chance, p ≈ .001). **B2.3** (best transformer by validation accuracy: time-patch, stop on accuracy; Integrated Gradients, Fig. 3): deleting the top-2/top-8 attributed 250 ms tokens gives 66.1/59.2% vs 64.9/61.5% for random tokens (weakly faithful). 41% (transformer) and 52% (EEGNet) of attribution falls in 0.25–1.5 s (31% if uniform). Zeroing the 17 frontal channels: transformer 53.8%, EEGNet 77.7%; sensorimotor: 67.2% / 76.6%. Regressing the HEOG proxies out of every channel (coefficients from training subjects; post-hoc): transformer 52.7%, EEGNet 65.0%. Per test subject (n = 10), transformer accuracy tracks A3 ocular accuracy (ρ = +0.64, p = 0.04), not mu LI (ρ = +0.04); EEGNet: ocular ρ = +0.45, LI ρ = +0.57 (p = 0.09). All 10 test subjects are A2-absent (random split), which limits the LI correlation. **Conclusion:** the transformer reads the A3 ocular confound and nothing else we can detect; EEGNet reads it for about half of its above-chance accuracy, with an unresolved remainder.

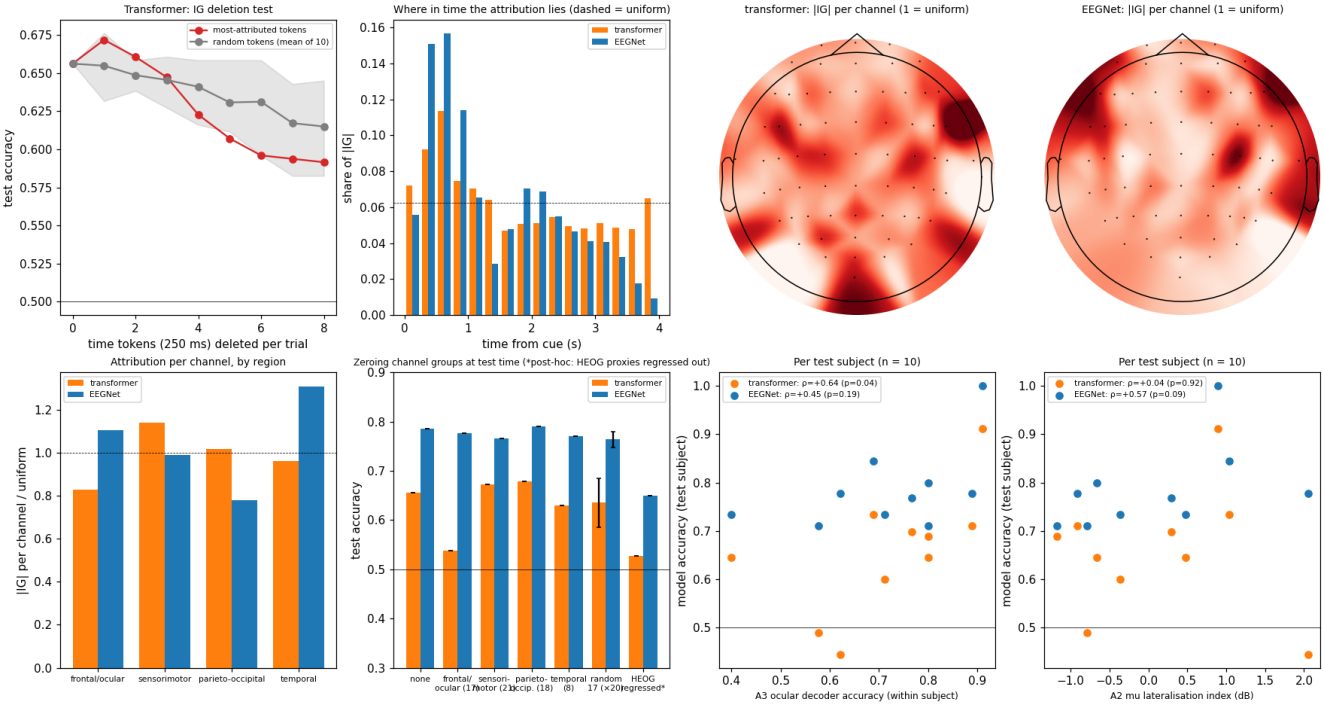


Fig. 3. B2.3: IG deletion test; attribution over time and per channel; per-region attribution; channel-group deletion and HEOG regression; per-test-subject accuracy vs A3 ocular accuracy and A2 mu LI.

Write-up

(a) If a model scores 70% on these subjects, how much of that could be explained without motor imagery? Potentially all of it. Eye movements towards the cued side alone decode left/right at 76% within subject (31/37 subjects above chance) and at **79.7%** across subjects — an LDA on three frontal bipolar derivations trained on the 23 training subjects and tested on the 10 test subjects, i.e. above 70% with no sensorimotor channel at all. The motor route on the same subjects reaches 52.7% within subject (8/37) and 48.0% across subjects; lateralised mu ERD is significant in only 5/37 subjects. The networks confirm it: removing the HEOG component at test time takes the transformer from 65.6% to 52.7% (chance) and EEGNet from 78.6% to 65.0%. So of a 70% score (20 points above chance), the ocular route can supply all 20; for our EEGNet, at least 14 of its 29 points above chance are ocular and the rest is unattributed. A credible motor-imagery claim on this dataset needs ocular control (fixation, EOG regression or ICA, or a window and channel set shown to carry no HEOG) and a comparison against this confound baseline, not against 50%.

(b) Mechanistic explanation of B2. *Scaling.* The decisive feature is one large, slow potential with the same polarity in nearly every subject. EEGNet's first two layers are a temporal filter followed by one spatial filter per temporal filter (depthwise weights, one per electrode): a single low-pass temporal filter plus a spatial filter ≈ F7–F8 is exactly the HEOG decoder, so ~225 trials from 5 subjects suffice and more subjects add nothing (76.8 → 78.6%). **This contradicted my prediction** of a rising EEGNet curve: I expected a cross-subject problem to need many subjects, but the confound is easier to share across people than ERD. The transformer has no such built-in structure — its spatial projection, temporal filters and attention are learned jointly — and it is at chance with 5 subjects, reaching 64.3% with 23. *Tokenisation.* Time-patch tokens encode electrode identity in the spatial-projection weights (one weight per electrode per virtual channel); channel tokens only through the learned identity embedding. Without it, a shared token encoder, attention and mean pooling make the model exactly invariant to permuting electrodes: it cannot represent any C3-vs-C4 or F7-vs-F8 contrast, so lateralised ERD is unusable. I predicted chance; **it scored 57.4%, contradicting me**, because the eye potential is not mirror-symmetric: during the saccade left-minus-right is +11.7 µV at F7 but -7.7 µV at F8, and the mean over all 64 channels shifts by +2.4 µV (32/37 subjects) — what an off-midline recording reference would produce, the same undocumented reference A2 had to work around. An order-blind model reads that global shift. *Reliance.* The transformer's evidence sits in the frontal channels (zeroing them: chance) and in the HEOG component (regressing it out: chance). EEGNet survives zeroing any one region because the eye potential volume-conducts to temporal (FT7/T7 as large as F7) and, through the reference, all channels; its depthwise spatial filters can pick it up anywhere, so only removing the component itself costs it 14 points. Token deletion was only weakly faithful because the 41-sample temporal filters straddle 40-sample token boundaries and the saccade spans several tokens.